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NEW: Visit the new Overleaf Learning Center to explore free online courses and live webinars — everything you need to master LaTeX and get the most out of Overleaf.

新增：访问全新的Overleaf 学习中心，探索免费在线课程和实时网络研讨会——这里有你掌握 LaTeX 并充分利用 Overleaf 所需的一切。

This introductory tutorial does not assume any prior experience of LaTeX but, hopefully, by the time you are finished, you will not only have written your first LaTeX document but also acquired sufficient knowledge and confidence to take the next steps toward LaTeX proficiency.

本入门教程不要求你具备任何 LaTeX 基础，但希望当你学完本教程时，不仅能完成你的第一篇 LaTeX 文档，还能掌握足够的知识、建立足够的信心，朝着熟练掌握 LaTeX 的目标迈出下一步。

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What is LaTeX? 什么是LaTeX?

LaTeX (pronounced "*LAY*-tek" or "*LAH*-tek") is a tool for typesetting professional-looking documents. However, LaTeX's mode of operation is quite different to many other document-production applications you may have used, such as Microsoft Word or LibreOffice Writer: those "WYSIWYG" tools provide users with an interactive page into which they type and edit

their text and apply various forms of styling. LaTeX works very differently: instead, your document is a plain text file interspersed with LaTeX *commands* used to express the desired (typeset) results. To produce a visible, typeset document, your LaTeX file is processed by a piece of software called a *TeX engine* which uses the commands embedded in your text file to guide and control the typesetting process, converting the LaTeX commands and document text into a professionally typeset PDF file. This means you only need to focus on the *content* of your document and the computer, via LaTeX commands and the TeX engine, will take care of the *visual appearance* (formatting).

LaTeX（发音为“*莱*-泰克”或“拉-泰克”）是一款用于排版出专业格式文档的工具。不过，LaTeX 的运行模式与你可能用过的许多其他文档制作软件（如 Microsoft Word 或 LibreOffice Writer）大相径庭：那些“所见即所得”工具会为用户提供一个交互式页面，用户可在其中输入、编辑文本并应用各类样式设置。LaTeX 的工作方式则截然不同：你的文档本质上是一个纯文本文件，其中穿插着 LaTeX *命*令，这些命令用于表达预期的（排版）效果。要生成可查看的排版文档，需通过名为 *TeX 引擎*的软件处理你的 LaTeX 文件；该引擎会利用文本文件中嵌入的命令来指导和控制排版流程，将 LaTeX 命令与文档文本转换为专业排版的 PDF 文件。这意味着你只需专注于文档的*内容*，而视觉呈现（格式设置）则由计算机通过 LaTeX 命令和 TeX 引擎来完成。

Why learn LaTeX? 为什么要学习LaTeX?

Various arguments can be proposed for, or against, learning to use LaTeX instead of other document-authoring applications; but, ultimately, it is a personal choice based on preferences, affinities, and documentation requirements.

人们可以提出各种支持或反对学习使用 LaTeX 而非其他文档编辑软件的论点；但归根结底，这是基于个人偏好、喜好和文档需求做出的自主选择。

Arguments in favour of LaTeX include:

支持使用LaTeX的理由包括：

- support for typesetting extremely complex mathematics, tables and technical content for the physical sciences;

  支持为物理科学领域排版极其复杂的数学公式、表格和技术内容

- facilities for footnotes, cross-referencing and management of bibliographies;

  提供脚注、交叉引用以及参考文献管理的功能；

- ease of producing complicated, or tedious, document elements such as indexes, glossaries, table of contents, lists of figures;

  轻松生成复杂或繁琐的文档元素，如索引、术语表、目录、图表列表；

- being highly customizable for bespoke document production due to its intrinsic programmability and extensibility through thousands of free add-on packages.

  凭借其内在的可编程性以及通过数千个免费附加组件实现的可扩展性，该工具可高度定制化，以满足定制化文档制作的需求。

Overall, LaTeX provides users with a great deal of control over the production of documents which are typeset to extremely high standards. Of course, there are types of documents or publications where LaTeX doesn't shine, including many "free form" page designs typically found in magazine-type publications.

总体而言，LaTeX 能让用户高度掌控文档的排版制作，使其达到极高的排版标准。当然，在某些类型的文档或出版物中，LaTeX 并不占优势，比如杂志类出版物中常见的许多"自由形式"页面设计。

One important benefit of LaTeX is the separation of document content from document style: once you have written the content of your document, its appearance can be changed with ease. Similarly, you can create a LaTeX file which defines the layout/style of a particular document type and that file can be used as a *template* to standardise authorship/production of additional documents of that type; for example, this allows scientific publishers to create article templates, in LaTeX, which authors use to write papers for submission to journals. Overleaf has a gallery containing thousands of templates, covering an enormous range of document types—everything from scientific articles, reports and books to CVs and presentations. Because these templates define the layout and style of the document, authors need only to open them in Overleaf—creating a new project—and

commence writing to add their content.

LaTeX 的一个重要优势是将文档内容与文档格式分离：一旦你编写完文档内容，就能轻松更改其外观。同样，你可以创建一个 LaTeX 文件来定义特定文档类型的版式/格式，该文件可作为*模板*，用于统一该类其他文档的撰写/制作；例如，这使得学术出版社能够用 LaTeX 制作文章模板，供作者撰写论文并投稿到期刊。Overleaf 设有包含数千个模板的图库，涵盖种类繁多的文档类型——从学术文章、报告、书籍到简历和演示文稿一应俱全。由于这些模板定义了文档的版式和格式，作者只需在 Overleaf 中打开它们——新建一个项目——即可开始撰写并添加自己的内容。

Writing your first piece of LaTeX 编写你的第一段 LaTeX 代码

The first step is to create a new LaTeX project. You can do this on your own computer by creating a new `.tex` file; alternatively, you can start a new project in Overleaf.

第一步是创建一个新的 LaTeX 项目。你可以在自己的电脑上通过新建一个 `.tex` 文件来完成；或者，你也可以 在 Overleaf 中新建一个项目。

Let's start with the simplest working example, which can be opened directly in Overleaf:

我们从最简单的可直接在 Overleaf 中打开的示例开始：

```
\documentclass{article}
\begin{document}
First document. This is a simple example, with
no
extra parameters or packages included.
\end{document}
```

```
\documentclass{article}\begin{document}第一个文档。这是一个简单的示例，没有包含任何额外的参数或宏包。
\end{document}
```

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：

First document. This is a simple example, with no extra parameters or packages included.

You can see that LaTeX has automatically indented the first line of the paragraph, taking care of that formatting for you. Let's have a closer look at what each part of our code does.

你可以看到，LaTeX 会自动缩进段落的首行，帮你完成了该格式设置。我们来仔细看看代码的各部分分别有什么作用。

The first line of code, `\documentclass{article}`, declares the document type known as its *class*, which controls the overall appearance of the document. Different types of documents require different classes; i.e., a CV/resume will require a different class than a scientific paper which might use the standard LaTeX `article` class. Other types of documents you may be working on may require different classes such as `book` or `report`. To get some idea of the many LaTeX class types available, visit the relevant page on CTAN (Comprehensive TeX Archive Network).

第一行代码 `\documentclass{article}` 声明了文档类型，即其*文档类*，它控制着文档的整体外观。不同类型的文档需要不同的文档类；例如，简历需要的文档类与可能使用标准 LaTeX `article` 文档类的学术论文不同。你可能要编写的其他类型文档可能需要 `book` 或 `report` 等不同的文档类。若想了解众多可用的 LaTeX 文档类类型，可访问 CTAN（TeX 综合档案网络）的相关页面。

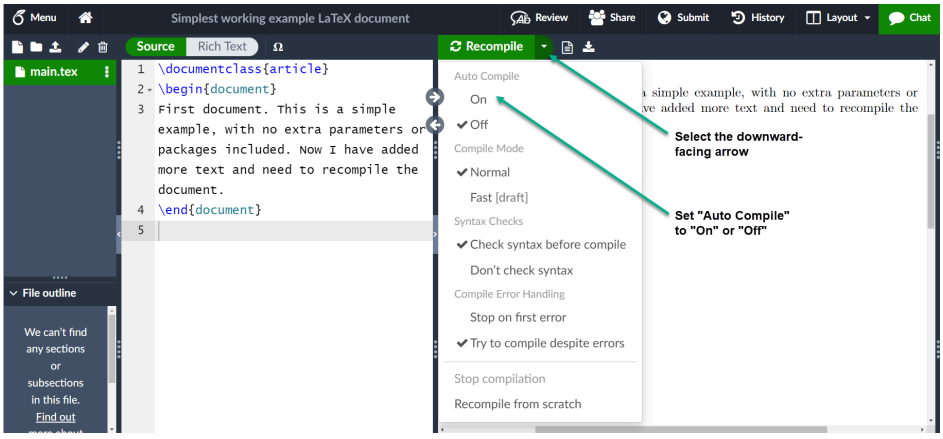
Having set the document class, our content, known as the *body* of the document, is written between the `\begin{document}` and `\end{document}` tags. After opening the example above, you can make changes to the text and, when finished, view the resulting typeset PDF by *recompiling the document*. To do this in Overleaf, simply hit **Recompile**, as demonstrated in this brief video clip:

设置完文档类之后，我们的内容（即文档的*正文*）需写在`\begin{document}`和`\end{document}`标签之间。打开上面的示例后，你可以修改文本，完成后通过*重新编译文档*来查看生成的排版好的PDF。在Overleaf中操作时，只需点击**重新编译**即可，这段简短的视频演示了具体操作：

0:00 / 0:15

Any Overleaf project can be configured to recompile automatically each time it is edited: click the small arrow next to the **Recompile** button and set **Auto Compile** to **On**, as shown in the following screengrab:

任何 Overleaf 项目都可配置为每次编辑时自动重新编译：点击**重新编译**按钮旁的小箭头，然后将**自动编译**设为**开启**，如下方截图所示：



Having seen how to add content to our document, the next step is to give it a title. To do this, we must talk briefly about the **preamble**.

了解了如何向文档添加内容后，下一步是为其添加标题。为此，我们需要简要谈谈**导言区**。

The preamble of a document
文档的导言区

The screengrab above shows Overleaf storing a LaTeX document as a file called `main.tex`: the `.tex` file extension is, by convention, used when naming files containing your document's LaTeX code.

上方的截图显示 Overleaf 将一个 LaTeX 文档存储为名为 `main.tex` 的文件：按照惯例，在命名包含文档 LaTeX 代码的文件时，会使用 `.tex` 文件扩展名。

The previous example showed how document content was entered after the `\begin{document}` command; however, everything in your `.tex` file appearing *before* that point is called the *preamble*, which acts as the document's "setup" section. Within the preamble you define the document class (type) together with specifics such as languages to be used when writing the document; loading *packages* you would like to use (more on this later), and it is where you'd apply other types of configuration.

上一个示例展示了文档内容是在`\begin{document}`命令之后输入的；不过，你的`.tex`文件中出现在*该位置之前*的所有内容都被称为*导言区*，它相当于文档的"设置"部分。在导言区中，你可以定义文档类（类型），以及编写文档时要使用的语言等具体信息；加载你希望使用的*宏包*（相关内容会在后续详细说明），同时也是在这里进行其他类型的配置。

A minimal document preamble might look like this:

一个最基础的文档导言区可能如下所示：

```
\documentclass[12pt, letterpaper]{article}
\usepackage{graphicx}
```

where `\documentclass[12pt, letterpaper]{article}` defines the overall class (type) of document. Additional parameters, which must be separated by commas, are included in square brackets (`[...]`) and used to configure this instance of the article class; i.e., settings we wish to use for this particular `article`-class-based document.

其中`\documentclass[12pt, letterpaper]{article}`定义了文档的整体类型。额外的参数（必须用逗号分隔）包含在方括号`[...]`中，用于配置article文档类的该实例；即我们希望此特定基于`article`类的文档所使用的设置。

In this example, the two parameters do the following:

在这个示例中，这两个参数的作用如下：

- `12pt` sets the font size `12pt` 设置字体大小
- `letterpaper` sets the paper size `letterpaper`设置纸张尺寸

Of course other font sizes, `9pt`, `11pt`, `12pt`, can be used, but if none is specified, the default size is `10pt`. As for the paper size, other possible values are `a4paper` and `legalpaper`. For

further information see the article about page size and margins.

当然也可以使用其他字号，`9pt`、`11pt`、`12pt`，但如果未指定具体字号，默认字号为`10pt`。至于纸张尺寸，其他可选值还有`a4paper`和`legalpaper`。更多信息请参阅关于页面尺寸与页边距的文章。

The preamble line 导言行

```
\usepackage{graphicx}
```

is an example of loading an external package (here, `graphicx`) to extend LaTeX's capabilities, enabling it to import external graphics files. LaTeX packages are discussed in the section Finding and using LaTeX packages.

这是加载外部宏包（此处为`graphicx`）以扩展 LaTeX 功能的示例，可让 LaTeX 导入外部图形文件。LaTeX 宏包的相关内容将在查找并使用 LaTeX 宏包一节中进行讲解。

Including title, author and date information

包含标题、作者和日期信息

Adding a title, author and date to our document requires three more lines in the *preamble* (*not* the main body of the document). Those lines are:

在我们的文档中添加标题、作者和日期需要在*导言区*（*非*文档的主体部分）再增加三行代码。这些行分别是：

- `\title{My first LaTeX document}`: the document title

  `\title{My first LaTeX document}`：文档标题

- `\author{Hubert Farnsworth}`: here you write the name of the author(s) and, optionally, the `\thanks` command within the curly braces:

  `\author{Hubert Farnsworth}`：在此处填写作者姓名，也可在花括号内添加`\thanks`命令：

  - `\thanks{Funded by the Overleaf team.}`: can be added after the name of the author, inside the

braces of the `author` command. It will add a superscript and a footnote with the text inside the braces. Useful if you need to thank an institution in your article.

`\thanks{Funded by the Overleaf team.}`：可添加在作者姓名之后、`author` 命令的花括号内。它会添加上标以及一个包含花括号内文本的脚注。若需在文章中致谢某机构，此命令十分实用。

- `\date{August 2022}`: you can enter the date manually or use the command `\today` to typeset the current date every time the document is compiled

  `\date{August 2022}`：你可以手动输入日期，或使用命令 `\today` 在每次编译文档时排版当前日期

With these lines added, your preamble should look something like this:

添加完这些代码后，你的导言区应该大致如下所示：

```
\documentclass[12pt, letterpaper]{article}
\title{My first LaTeX document}
\author{Hubert Farnsworth\thanks{Funded by the
Overleaf team.}}
\date{August 2022}
```

```
\documentclass[12pt, letterpaper]
{article}\title{我的第一篇 LaTeX 文档}\author{休伯
特·法恩斯沃思\thanks{由 Overleaf 团队资
助。}}\date{2022年8月}
```

To typeset the title, author and date use the `\maketitle` command within the *body* of the document:

要排版标题、作者和日期，请在文档的*正文*中使用`\maketitle`命令：

```
\begin{document}
\maketitle
We have now added a title, author and date to
our first \LaTeX{} document!
\end{document}
```

```
\begin{document}\maketitle我们已经在第一个
\LaTeX{} 文档中添加了标题、作者和日期!
\end{document}
```

The preamble and body can now be combined to produce a complete document which can be opened in Overleaf:

现在可以将导言和正文合并，生成一份可在 Overleaf 中打开的完整文档：

```
\documentclass[12pt, letterpaper]{article}
\title{My first LaTeX document}
\author{Hubert Farnsworth\thanks{Funded by the
Overleaf team.}}
\date{August 2022}
\begin{document}
\maketitle
We have now added a title, author and date to
our first \LaTeX{} document!
\end{document}
```

```
\documentclass[12pt, letterpaper]
{article}\title{我的第一篇 LaTeX 文档}\author{休伯
特·法恩斯沃思\thanks{由 Overleaf 团队资
助。}}\date{2022年8月}\begin{document}\maketitle
我们已经为第一篇 \LaTeX{} 文档添加了标题、作者和日期信
息！\end{document}
```

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：

My first LaTeX document

Hubert Farnsworth*

August 2022

We have now added a title, author and date to our first LaTeX document!

*Funded by the Overleaf team.

1

Adding comments 添加注释

LaTeX is a form of “program code”, but one which specializes in document typesetting; consequently, as with code written in any other programming language, it can be very useful to include comments within your document. A LaTeX comment is a section of text that will not be typeset or affect the document in any way—often used to add “to do” notes; include explanatory notes; provide in-line explanations of tricky macros or comment-out lines/sections of LaTeX code when debugging.

LaTeX 是一种“程序代码”形式，但它专门用于文档排版；因此，与用其他任何编程语言编写的代码一样，在文档中添加注释会非常有用。LaTeX 注释是一段不会被排版或以任何方式影响文档的文本

——通常用于添加“待办”笔记、说明性注释、对复杂宏进行行内解释，或在调试时注释掉 LaTeX 代码的行或部分内容。

To make a comment in LaTeX, simply write a % symbol at the beginning of the line, as shown in the following code which uses the example above:

要在 LaTeX 中添加注释，只需在行首写一个%符号，如下所示的代码就使用了上面的示例：

```
\documentclass[12pt, letterpaper]{article}
\title{My first LaTeX document}
\author{Hubert Farnsworth\thanks{Funded by the
Overleaf team.}}
\date{August 2022}
\begin{document}
\maketitle
We have now added a title, author and date to
our first \LaTeX{} document!

% This line here is a comment. It will not be
typeset in the document.
\end{document}
```

```
\documentclass[12pt, letterpaper]
{article}\title{我的第一篇 LaTeX 文档}\author{休伯特·法恩斯沃思\thanks{由 Overleaf 团队资助。}}\date{2022年8月}\begin{document}\maketitle
我们已经为我们的第一篇 \LaTeX{} 文档添加了标题、作者和日期！% 这一行是注释，不会在文档中排版。
\end{document}
```

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces output that is identical to the previous LaTeX code which did not contain the comment.

此示例生成的输出与之前不含注释的 LaTeX 代码完全相同。

Bold, italics and underlining
粗体、斜体和下划线

Next, we will now look at some text formatting commands:

接下来，我们来看看一些文本格式命令：

- **Bold**: bold text in LaTeX is typeset using the `\textbf{...}` command.

  粗体：LaTeX 中的粗体文本通过`\textbf{...}`命令排版。

- *Italics*: italicised text is produced using the `\textit{...}` command.

  斜体：斜体文本通过`\textit{...}`命令生成。

- Underline: to underline text use the `\underline{...}` command.

  下划线：要给文本添加下划线，请使用 `\underline{...}` 命令。

The next example demonstrates these commands:

下一个示例演示了这些命令：

```
Some of the \textbf{greatest}
discoveries in \underline{science}
were made by \textbf{\textit{accident}}.
```

科学领域的一些**最伟大**发现是**意外**做出的。

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：

Some of the **greatest** discoveries in science were made by ***accident***.

Another very useful command is `\emph{argument}`, whose effect on its *argument* depends on the context. Inside normal text, the emphasized text is italicized, but this behaviour is reversed if used inside an italicized text—see the next example:

另一个非常有用的命令是 `\emph{argument}`，它对其 *argument* 的效果取决于上下文。在普通文本中，强调的文本会显示为斜体，但如果在斜体文本中使用该命令，效果会相反——请看下面的示例：

```
Some of the greatest \emph{discoveries} in
science
were made by accident.
```

```
\textit{Some of the greatest \emph{discoveries}
in science were made by accident.}

\textbf{Some of the greatest \emph{discoveries}
in science were made by accident.}
```

科学史上一些最伟大的**发现**都是偶然做出的。*科学史上一些最伟大的**发现**都是偶然做出的。***科学史上一些最伟大的**发现**都是偶然做出的。**

Open this **\emph** example in Overleaf.

在 Overleaf 中打开这个 \emph 示例。

This example produces the following output:

该示例生成以下输出：

Some of the greatest *discoveries* in science were made by accident.
Some of the greatest discoveries *in science were made by accident.*
Some of the greatest *discoveries* in science were made by accident.

- **Note:** some packages, such as Beamer, change the behaviour of the \emph command.

  注：部分宏包（如 Beamer）会更改 \emph 命令的行为。

Adding images 添加图片

In this section we will look at how to add images to a LaTeX document. Overleaf supports three ways to insert images:

在本节中，我们将探讨如何向 LaTeX 文档中添加图片。Overleaf 支持三种插入图片的方式：

1. Use the **Insert Figure** button(), located on the editor toolbar, to insert an image into **Visual Editor** or **Code Editor**.

   使用位于编辑器工具栏上的**插入图片**按钮()，在**可视化编辑器**或**代码编辑器**中插入图片。

2. Copy and paste an image into **Visual Editor** or **Code Editor**.

   复制并粘贴图片到**可视化编辑器**或**代码编辑器**中。

3. Use **Code Editor** to write LaTeX code that inserts a graphic.

使用代码编辑器编写用于插入图形的 LaTeX 代码。

Options 1 and 2 automatically generate the LaTeX code required to insert images, but here we introduce option 3—note that you will need to upload those images to your Overleaf project. The following example demonstrates how to include a picture:

选项1和选项2会自动生成插入图片所需的LaTeX代码，下面我们介绍选项3——请注意，你需要将这些图片上传到你的Overleaf项目中。以下示例演示了如何插入一张图片：

```
\documentclass{article}
\usepackage{graphicx} %LaTeX package to import
graphics
\graphicspath{{images/}} %configuring the
graphicx package

\begin{document}
The universe is immense and it seems to be
homogeneous,
on a large scale, everywhere we look.

% The \includegraphics command is
% provided (implemented) by the
% graphicx package
\includegraphics{universe}

There's a picture of a galaxy above.
\end{document}
```

```
\documentclass{article}\usepackage{graphicx} %
用于导入图形的LaTeX宏包\graphicspath{{images/}} %
配置graphicx宏包\begin{document}宇宙浩瀚无垠，从宏观
尺度来看，它似乎处处都具有均匀性。% \includegraphics
命令由% graphicx宏包%提供（实现）
\includegraphics{universe} 上方是一张星系的图片。
\end{document}
```

Open this image example in Overleaf.

在 Overleaf 中打开这个图片示例。

This example produces the following output:

该示例生成以下输出：

The universe is immense and it seems to be homogeneous, on a large scale, everywhere we look.



There's a picture of a galaxy above.

Importing graphics into a LaTeX document needs an add-on *package* which provides the commands and features required to include external graphics files. The above example loads the `graphicx` package which, among many other commands, provides `\includegraphics{...}` to import graphics and `\graphicspath{...}` to advise LaTeX where the graphics are located.

在 LaTeX 文档中导入图形需要一个附加的*宏包*，该宏包提供了插入外部图形文件所需的命令和功能。上述示例加载了`graphicx`宏包，它除了众多其他命令外，还提供了用于导入图形的`\includegraphics{...}`，以及用于告知 LaTeX 图形文件存放位置的`\graphicspath{...}`。

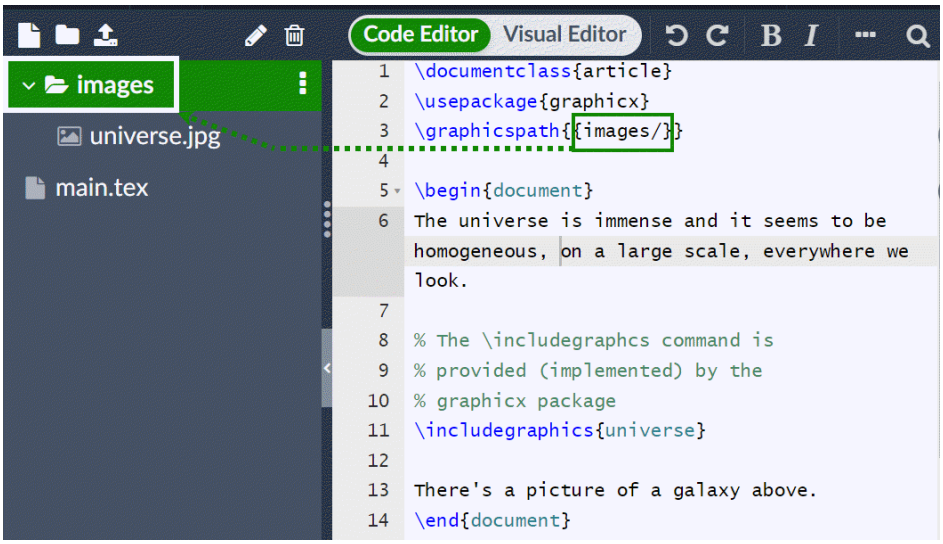
To use the `graphicx` package, include the following line in your Overleaf document preamble:

要使用 `graphicx` 宏包，请在 Overleaf 文档的导言区中加入以下代码行：

```
\usepackage{graphicx}
```

In our example the command `\graphicspath{{images/}}` informs LaTeX that images are kept in a folder named `images`, which is contained in the current directory:

在我们的示例中，命令 `\graphicspath{{images/}}` 会告知 LaTeX 图像存放在当前目录下名为 `images` 的文件夹中：



The `\includegraphics{universe}` command does the actual work of inserting the image in the document. Here, `universe` is the name of the image file but without its extension.

`\includegraphics{universe}` 命令负责将图片插入文档的实际操作。此处 `universe` 是图片文件的名称，但不包含其扩展名。

Note: 注意：

- Although the full file name, including its extension, is allowed in the `\includegraphics` command, it's considered best practice to omit the file extension because it will prompt LaTeX to search for all the supported formats.

  尽管在`\includegraphics`命令中允许使用包含扩展名的完整文件名，但最佳做法是省略文件扩展名，因为这样会促使LaTeX搜索所有受支持的格式。

- Generally, the graphic's file name should not contain white spaces or multiple dots; it is also recommended to use lowercase letters for the file extension when uploading image files to Overleaf.

  通常来说，图片的文件名不应包含空格或多个点；将图片文件上传到 Overleaf 时，也建议使用小写字母作为文件扩展名。

More information on LaTeX packages can be found at the end of this tutorial in the section Finding and using LaTeX packages.

有关 LaTeX 宏包的更多信息可在本教程末尾的 查找和使用 LaTeX 宏包一节中找到。

Captions, labels and references
标题、标签与引用

Images can be captioned, labelled and referenced by means of the `figure` environment, as shown below:

可以通过`figure`环境为图片添加标题、标注和引用，如下所示：

```
\documentclass{article}
\usepackage{graphicx}
\graphicspath{{images/}}

\begin{document}

\begin{figure}[h]
    \centering
    \includegraphics[width=0.75\textwidth]
{mesh}
    \caption{A nice plot.}
    \label{fig:mesh1}
\end{figure}

As you can see in figure \ref{fig:mesh1}, the
function grows near the origin. This example is
on page \pageref{fig:mesh1}.

\end{document}
```

```
\documentclass{article}\usepackage{graphicx}\gr
aphicspath{{images/}}\begin{document}\begin{fig
ure}
[h]\centering\includegraphics[width=0.75\textwi
dth]{mesh}\caption{一张精美的图
表。}\label{fig:mesh1}\end{figure}如图
\ref{fig:mesh1} 所示，该函数在原点附近呈增长趋势。本
示例位于 \pageref{fig:mesh1} 页。\end{document}
```

Open this image example in Overleaf.

在 Overleaf 中打开此图片示例。

This example produces the following output:

该示例生成以下输出：

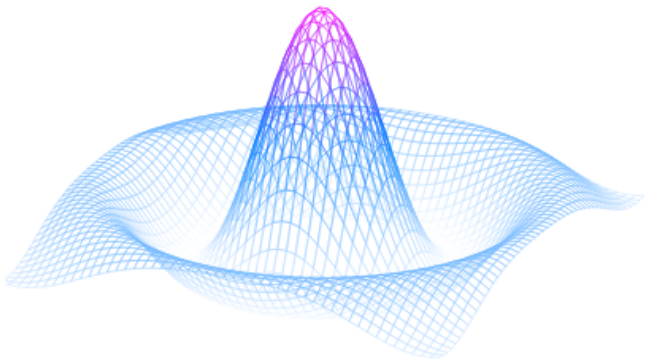


Figure 1: A nice plot.

As you can see in figure 1, the function grows near the origin. This example is on page 1.

There are several noteworthy commands in the example:

示例中有几个值得注意的命令：

- **`\includegraphics[width=0.75\textwidth]{mesh}`**: This form of `\includegraphics` instructs LaTeX to set the figure's width to 75% of the text width—whose value is stored in the `\textwidth` command.

  `\includegraphics[width=0.75\textwidth]{mesh}`：这种形式的`\includegraphics`命令指示LaTeX将图片宽度设置为文本宽度的75%——文本宽度的值存储在`\textwidth`命令中。

- **`\caption{A nice plot.}`**: As its name suggests, this command sets the figure caption which can be placed above or below the figure. If you create a list of figures this caption will be used in that list.

  `\caption{A nice plot.}`：顾名思义，该命令用于设置图表标题，标题可置于图表上方或下方。若你创建图表目录，该标题将被用于目录中。

- **`\label{fig:mesh1}`**: To reference this image within your document you give it a label using the `\label` command. The label is used to generate a number for the image and, combined with the next command, will allow you to reference it.

  `\label{fig:mesh1}`：要在文档中引用此图片，需使用`\label`命令为其指定标签。该标签用于生成图片编号，结合下一个命令，你就可以对其进行引用。

- **`\ref{fig:mesh1}`**: This code will be substituted by the number corresponding to the referenced figure.

  `\ref{fig:mesh1}`：此代码将被替换为所引用图形对应的编号。

Images incorporated in a LaTeX document should be placed inside a `figure` environment, or similar, so that LaTeX can automatically position the image at a suitable location in your document.

LaTeX 文档中插入的图片应放在 `figure` 环境或类似环境中，这样 LaTeX 就能自动将图片放置在文档的合适位置。

Further guidance is contained in the following Overleaf help articles:

以下 Overleaf 帮助文章包含更多指导：

- Positioning of Figures 图形的排版
- Inserting Images 插入图片

Creating lists in LaTeX
在 LaTeX 中创建列表

You can create different types of list using *environments*, which are used to encapsulate the LaTeX code required to implement a specific typesetting feature. An environment starts with `\begin{`*`environment-name`*`}` and ends with `\end{`*`environment-name`*`}` where *`environment-name`* might be `figure`, `tabular` or one of the list types: `itemize` for unordered lists or `enumerate` for ordered lists.

你可以使用*环境*创建不同类型的列表，环境用于封装实现特定排版功能所需的 LaTeX 代码。环境以`\begin{environment-name}`开头，以`\end{environment-name}`结尾，其中`environment-name`可以是`figure`、`tabular`，或是以下列表类型之一：用于无序列表的`itemize`，或用于有序列表的`enumerate`。

Unordered lists 无序列表

Unordered lists are produced by the `itemize` environment. Each list entry must be preceded by the `\item` command, as

shown below:

无序列表由`itemize`环境生成。每个列表项前都必须加上`\item`命令，如下所示：

```
\documentclass{article}
\begin{document}
\begin{itemize}
  \item The individual entries are indicated
with a black dot, a so-called bullet.
  \item The text in the entries may be of any
length.
\end{itemize}
\end{document}
```

```
\documentclass{article}\begin{document}\begin{i
temize}\item 各个条目用黑色圆点表示，即所谓的项目符
号。\item 条目的文本可以是任意长度。
\end{itemize}\end{document}
```

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：

- The individual entries are indicated with a black dot, a so-called bullet.
- The text in the entries may be of any length.

You can also open this larger Overleaf project which demonstrates various types of LaTeX list.

你也可以打开这个更大的 Overleaf 项目，它展示了各种类型的 LaTeX 列表。

Ordered lists 有序列表

Ordered lists use the same syntax as unordered lists but are created using the `enumerate` environment:

有序列表使用与无序列表相同的语法，但需通过`enumerate`环境创建：

```
\documentclass{article}
\begin{document}
\begin{enumerate}
```

```
  \item This is the first entry in our list.
  \item The list numbers increase with each
entry we add.
\end{enumerate}
\end{document}
```

```
\documentclass{article}\begin{document}\begin{e
numerate}\item 这是我们列表中的第一项。\item 随着我
们添加每一项，列表编号依次递增。
\end{enumerate}\end{document}
```

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：

1. This is the first entry in our list.
2. The list numbers increase with each entry we add.

As with `unordered` lists, each entry must be preceded by the `\item` command which, here, automatically generates the numeric ordered-list label value, starting at 1.

与`unordered`一样，每个条目前都必须加上`\item`命令，该命令会在此处自动生成从1开始的数字有序列表标签值。

For further information you can open this larger Overleaf project which demonstrates various types of LaTeX list or visit our dedicated help article on LaTeX lists, which provides many more examples and shows how to create customized lists.

如需了解更多信息，你可以打开这个 更详细的 Overleaf 项目，它展示了各种类型的 LaTeX 列表；也可以访问我们专门的 LaTeX 列表帮助文章，其中提供了更多示例，并讲解了如何创建自定义列表。

Adding math to LaTeX
向 LaTeX 中添加数学公式

One of the main advantages of LaTeX is the ease with which mathematical expressions can be written. LaTeX provides two writing modes for typesetting mathematics:

LaTeX 的主要优势之一是能够轻松编写数学表达式。LaTeX 提供了两种排版数学公式的书写模式：

- *inline* math mode used for writing formulas that are part of a paragraph

  *行内*数学模式，用于编写属于段落的公式

- *display* math mode used to write expressions that are not part of a text or paragraph and are typeset on separate lines

  *显示*数学模式用于编写不属于文本或段落的表达式，这些表达式会单独排版成行

Inline math mode 行内数学模式

Let's see an example of inline math mode:

我们来看一个行内数学模式的示例：

```
\documentclass[12pt, letterpaper]{article}
\begin{document}
In physics, the mass-energy equivalence is
stated
by the equation $E=mc^2$, discovered in 1905 by
Albert Einstein.
\end{document}
```

```
\documentclass[12pt, letterpaper]
{article}\begin{document}在物理学中，质能等价关系由方程 $E=mc^2$ 表述，该方程由阿尔伯特·爱因斯坦于1905年发现。\end{document}
```

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：

In physics, the mass-energy equivalence is stated by the equation $E = mc^2$, discovered in 1905 by Albert Einstein.

To typeset inline-mode math you can use one of these delimiter pairs: `\( ... \)`, `$ ... $` or `\begin{math} ... \end{math}`, as demonstrated in the following example:

要排版行内数学公式，你可以使用以下这些定界符对之一：`\( ... \)`、`$ ... $` 或 `\begin{math} ... \end{math}`，如下例所示：

```
\documentclass[12pt, letterpaper]{article}
\begin{document}
\begin{math}
E=mc^2
\end{math} is typeset in a paragraph using
inline math mode---as is $E=mc^2$, and so too
is \(E=mc^2\).
\end{document}
```

```
\documentclass[12pt, letterpaper]
{article}\begin{document}\begin{math}E=mc^2\end
{math} 是在行内数学模式中排版的段落—$E=mc^2$ 也是如
此，\(E=mc^2\) 同样是。\end{document}
```

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：

$E = mc^2$ is typeset in a paragraph using inline math mode—as is $E = mc^2$, and so too is $E = mc^2$.

Display math mode 显示数学模式

Equations typeset in display mode can be numbered or unnumbered, as in the following example:

以显示模式排版的公式可以带编号或不带编号，如下例所示：

```
\documentclass[12pt, letterpaper]{article}
\begin{document}
The mass-energy equivalence is described by the
famous equation
\[ E=mc^2 \] discovered in 1905 by Albert
Einstein.

In natural units ($c = 1$), the formula
expresses the identity
\begin{equation}
E=m
\end{equation}
\end{document}
```

```
\documentclass[12pt, letterpaper]
{article}\begin{document}质能等价性由著名的方程\[
```

```
E=mc^2 \]描述，该方程由阿尔伯特·爱因斯坦于1905年提出。
在自然单位制（$c = 1$）下，该公式可表示为恒等式
\begin{equation}E=m\end{equation}\end{document}
```

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：

The mass-energy equivalence is described by the famous equation

$$E = mc^2$$

discovered in 1905 by Albert Einstein.

In natural units ($c = 1$), the formula expresses the identity

$$E = m \tag{1}$$

To typeset display-mode math you can use one of these delimiter pairs: `\[ ... \]`, `\begin{displaymath} ... \end{displaymath}` or `\begin{equation} ... \end{equation}`. Historically, typesetting display-mode math required use of `$$` characters delimiters, as in `$$ ... display math here ...$$`, but this method is no longer recommended: use LaTeX's delimiters `\[ ... \]` instead.

要排版显示模式数学公式，可使用以下分隔符对：`\[ ... \]`、`\begin{displaymath} ... \end{displaymath}` 或 `\begin{equation} ... \end{equation}`。过去，排版显示模式数学公式需要使用 `$` 字符分隔符，如 `$ ... display math here ...$`，但该方法 不再推荐使用，请改用 LaTeX 的分隔符 `\[ ... \]`。

More complete examples 更完整的示例

The following examples demonstrate a range of mathematical content typeset using LaTeX.

以下示例展示了使用 LaTeX 排版的各类数学内容。

```
\documentclass{article}
\begin{document}
Subscripts in math mode are written as $a_b$
and superscripts are written as $a^b$. These
can be combined and nested to write expressions
such as
```

```
\[ T^{i_1 i_2 \dots i_p}_{j_1 j_2 \dots j_q} =
T(x^{i_1},\dots,x^{i_p},e_{j_1},\dots,e_{j_q})
\]

We write integrals using $\int$ and fractions
using $\frac{a}{b}$. Limits are placed on
integrals using superscripts and subscripts:

\[ \int_0^1 \frac{dx}{e^x} =  \frac{e-1}{e} \]

Lower case Greek letters are written as
$\omega$ $\delta$ etc. while upper case Greek
letters are written as $\Omega$ $\Delta$.

Mathematical operators are prefixed with a
backslash as $\sin(\beta)$, $\cos(\alpha)$,
$\log(x)$ etc.
\end{document}
```

```
\documentclass{article}\begin{document}数学模式中的下标写作 $a_b$，上标写作 $a^b$。这些可以组合和嵌套来书写表达式，例如\[ T^{i_1 i_2 \dots i_p}_{j_1 j_2 \dots j_q} =
T(x^{i_1},\dots,x^{i_p},e_{j_1},\dots,e_{j_q})
\]我们使用 $\int$ 表示积分，使用 $\frac{a}{b}$ 表示分数。极限通过上标和下标标注在积分上：\[ \int_0^1 \frac{dx}{e^x} =  \frac{e-1}{e} \]小写希腊字母写作 $\omega$ $\delta$ 等，而大写希腊字母写作 $\Omega$ $\Delta$。数学运算符以反斜杠为前缀，例如 $\sin(\beta)$、$\cos(\alpha)$、$\log(x)$ 等。
\end{document}
```

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：

Subscripts in math mode are written as a_b and superscripts are written as a^b. These can be combined and nested to write expressions such as

$$T^{i_1 i_2 \ldots i_p}_{j_1 j_2 \ldots j_q} = T(x^{i_1}, \ldots, x^{i_p}, e_{j_1}, \ldots, e_{j_q})$$

We write integrals using $\int$ and fractions using $\frac{a}{b}$. Limits are placed on integrals using superscripts and subscripts:

$$\int_0^1 \frac{dx}{e^x} = \frac{e-1}{e}$$

Lower case Greek letters are written as ω δ etc. while upper case Greek letters are written as Ω Δ.

Mathematical operators are prefixed with a backslash as $\sin(\beta)$, $\cos(\alpha)$, $\log(x)$ etc.

The next example uses the `equation*` environment which is provided by the `amsmath` package, so we need to add the following line to our document preamble:

下一个示例使用了`equation*`环境，该环境由`amsmath`宏包提供，因此我们需要在文档导言区添加以下代码行：

```
\usepackage{amsmath}% For the equation*
environment
\usepackage{amsmath}% 用于 equation* 环境
```

For further information on using `amsmath` see our help article.

有关使用`amsmath`的更多信息，请参阅我们的帮助文章。

```
\documentclass{article}
\usepackage{amsmath}% For the equation*
environment
\begin{document}
\section{First example}

The well-known Pythagorean theorem \(x^2 + y^2
= z^2\) was proved to be invalid for other
exponents, meaning the next equation has no
integer solutions for \(n>2\):

\[ x^n + y^n = z^n \]

\section{Second example}

This is a simple math expression \
```

```
(\sqrt{x^2+1}\) inside text.
And this is also the same:
\begin{math}
\sqrt{x^2+1}
\end{math}
but by using another command.

This is a simple math expression without
numbering
\[\sqrt{x^2+1}\]
separated from text.

This is also the same:
\begin{displaymath}
\sqrt{x^2+1}
\end{displaymath}

\ldots and this:
\begin{equation*}
\sqrt{x^2+1}
\end{equation*}
\end{document}
```

```
\documentclass{article}\usepackage{amsmath}% 用于 equation* 环境\begin{document}\section{第一个示例}著名的勾股定理 \(x^2 + y^2 = z^2\) 被证明对其他指数不成立，这意味着下式对于 \(n>2\) 没有整数解：\[ x^n + y^n = z^n \]\section{第二个示例}这是文本内的一个简单数学表达式 \(\sqrt{x^2+1}\)。这也是一样的：\begin{math}\sqrt{x^2+1}\end{math}但使用了另一个命令。这是一个不带编号的简单数学表达式\[ \sqrt{x^2+1} \]与文本分隔开。这也是一样的：\begin{displaymath}\sqrt{x^2+1}\end{displaymath}……以及这个：\begin{equation*}\sqrt{x^2+1}\end{equation*}\end{document}
```

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：

1 First example

The well-known Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents, meaning the next equation has no integer solutions for $n > 2$:

$$x^n + y^n = z^n$$

2 Second example

This is a simple math expression $\sqrt{x^2 + 1}$ inside text. And this is also the same: $\sqrt{x^2 + 1}$ but by using another command.

This is a simple math expression without numbering

$$\sqrt{x^2 + 1}$$

separated from text.

This is also the same:

$$\sqrt{x^2 + 1}$$

. . . and this:

$$\sqrt{x^2 + 1}$$

The possibilities with math in LaTeX are endless so be sure to visit our help pages for advice and examples on specific topics:

在 LaTeX 中使用数学的可能性无穷无尽，因此请务必访问我们的帮助页面，获取特定主题的建议和示例：

- Mathematical expressions 数学表达式
- Subscripts and superscripts 下标和上标
- Brackets and Parentheses 方括号与圆括号
- Fractions and Binomials 分式与二项式
- Aligning Equations 方程对齐
- Operators 运算符
- Spacing in math mode 数学模式中的间距
- Integrals, sums and limits 积分、求和与极限
- Display style in math mode 数学模式中的显示样式
- List of Greek letters and math symbols

  希腊字母与数学符号列表
- Mathematical fonts 数学字体

Basic document structure 基础文档结构

Next, we explore abstracts and how to partition a LaTeX document into different chapters, sections and paragraphs.

接下来，我们将探讨摘要，以及如何将 LaTeX 文档划分为不同的章节、小节和段落。

Abstracts 摘要

Scientific articles usually provide an *abstract* which is a brief overview/summary of their core topics, or arguments. The next example demonstrates typesetting an abstract using LaTeX's `abstract` environment:

学术论文通常会提供一篇*摘要*，对其核心主题或论点进行简要概述/总结。下例演示了如何使用 LaTeX 的`abstract`环境排版摘要：

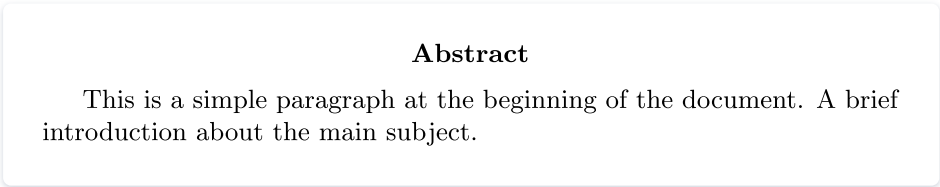
```
\documentclass{article}
\begin{document}
\begin{abstract}
This is a simple paragraph at the beginning of
the
document. A brief introduction about the main
subject.
\end{abstract}
\end{document}
```

```
\documentclass{article}\begin{document}\begin{a
bstract}这是文档开头的一个简单段落。关于主题的简要介
绍。\end{abstract}\end{document}
```

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：



Paragraphs and new lines 段落与换行

With the abstract in place, we can begin writing our first paragraph. The next example demonstrates:

完成摘要后，我们就可以开始撰写第一段了。下一个示例将演示：

- how a new paragraph is created by pressing the "enter" key twice, ending the current line and inserting a

subsequent blank line;

通过两次按下“Enter”键来创建新段落，这会结束当前行并插入一个后续的空行；

- how to start a new line without starting a new paragraph by inserting a manual line break using the `\\` command, which is a double backslash; alternatively, use the `\newline` command.

  如何通过插入`\\`命令（即双反斜杠）手动换行来开始新的一行而不新建段落；或者使用`\newline`命令。

The third paragraph in this example demonstrates use of the commands `\\` and `\newline`:

此示例中的第三段演示了命令 `\\` 和 `\newline` 的用法：

```
\documentclass{article}
\begin{document}

\begin{abstract}
This is a simple paragraph at the beginning of
the
document. A brief introduction about the main
subject.
\end{abstract}

After our abstract we can begin the first
paragraph, then press ``enter'' twice to start
the second one.

This line will start a second paragraph.

I will start the third paragraph and then add
\\ a manual line break which causes this text
to start on a new line but remains part of the
same paragraph. Alternatively, I can use the
\verb|\newline|\newline command to start a new
line, which is also part of the same paragraph.
\end{document}
```

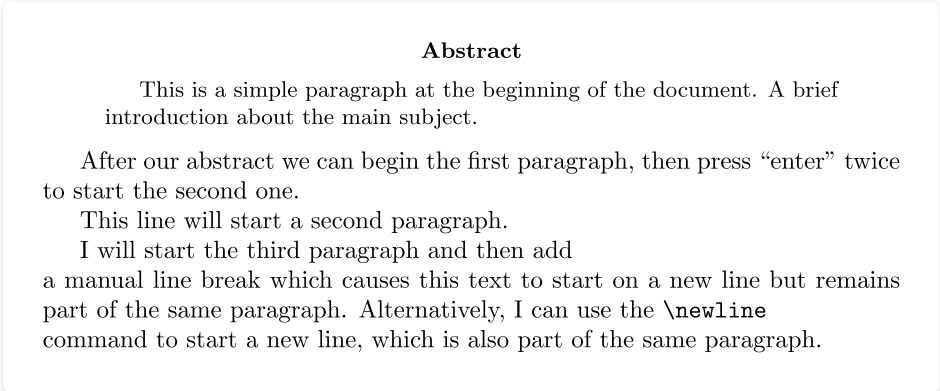
`\documentclass{article}\begin{document}\begin{abstract}`这是文档开头的一个简单段落。对主题的简要介绍。`\end{abstract}`摘要之后，我们可以开始第一段，然后按两次“回车”来启动第二段。这一行将开启第二段。我将开始第三段，然后添加`\\`手动换行符，这会使该文本从新的一行开始，但

仍属于同一段落。或者，我也可以使用
\verb|\newline|\newline命令来开启新的一行，这同样属于同一段落。\end{document}

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：



Abstract

This is a simple paragraph at the beginning of the document. A brief introduction about the main subject.

After our abstract we can begin the first paragraph, then press "enter" twice to start the second one.

This line will start a second paragraph.

I will start the third paragraph and then add
a manual line break which causes this text to start on a new line but remains part of the same paragraph. Alternatively, I can use the `\newline` command to start a new line, which is also part of the same paragraph.

Note how LaTeX automatically indents paragraphs—except immediately after document headings such as section and subsection—as we will see.

注意 LaTeX 会自动缩进段落——除了紧跟在章节、小节等文档标题之后的段落——我们稍后会看到。

New users are advised that multiple \\ or \newlines should not be used to "simulate" paragraphs with larger spacing between them because this can interfere with LaTeX's typesetting algorithms. The recommended method is to continue using blank lines for creating new paragraphs, without any \\, and load the parskip package by adding \usepackage{parskip} to the preamble.

新用户请注意，不要使用多个\\或\newline来"模拟"间距更大的段落，因为这会干扰LaTeX的排版算法。推荐的方法是继续使用空行来创建新段落，不要使用任何\\，并通过在导言区添加\usepackage{parskip}来加载parskip宏包。

Further information on paragraphs can be found in the following articles:

有关段落的更多信息可在以下文章中找到：

- Paragraphs and new lines 段落与换行
- How to change paragraph spacing in LaTeX

  如何在 LaTeX 中更改段落间距

- LaTeX Error: There's no line here to end provides additional advice and guidance on using `\\`.

  LaTeX 错误：此处没有可结束的行 提供了关于使用 `\\` 的额外建议和指导。

Chapters and sections 章节与小节

Longer documents, irrespective of authoring software, are usually partitioned into parts, chapters, sections, subsections and so forth. LaTeX also provides document-structuring commands but the available commands, and their implementations (what they do), can depend on the document class being used. By way of example, documents created using the `book` class can be split into parts, chapters, sections, subsections and so forth but the `letter` class does not provide (support) any commands to do that.

无论使用何种编辑软件，较长的文档通常都会被划分为部分、章、节、小节等层级。LaTeX 也提供了文档结构命令，但可用的命令及其具体实现（即它们的功能）可能取决于所使用的文档类。例如，使用`book`类创建的文档可分为部分、章、节、小节等，但`letter`类不提供任何相关命令来实现这一点。

This next example demonstrates commands used to structure a document based on the `book` class:

下一个示例演示了用于基于`book`类来构建文档的命令：

```
\documentclass{book}
\begin{document}

\chapter{First Chapter}

\section{Introduction}

This is the first section.

Lorem ipsum dolor sit amet, consectetuer
adipiscing
elit. Etiam lobortisfacilisis sem. Nullam nec
mi et
neque pharetra sollicitudin. Praesent
imperdietmi nec ante.
Donec ullamcorper, felis non sodales...
```

```
\section{Second Section}

Lorem ipsum dolor sit amet, consectetuer
adipiscing elit.
Etiam lobortis facilisissem.  Nullam nec mi et
neque pharetra
sollicitudin.  Praesent imperdiet mi necante...

\subsection{First Subsection}
Praesent imperdietmi nec ante. Donec
ullamcorper, felis non sodales...

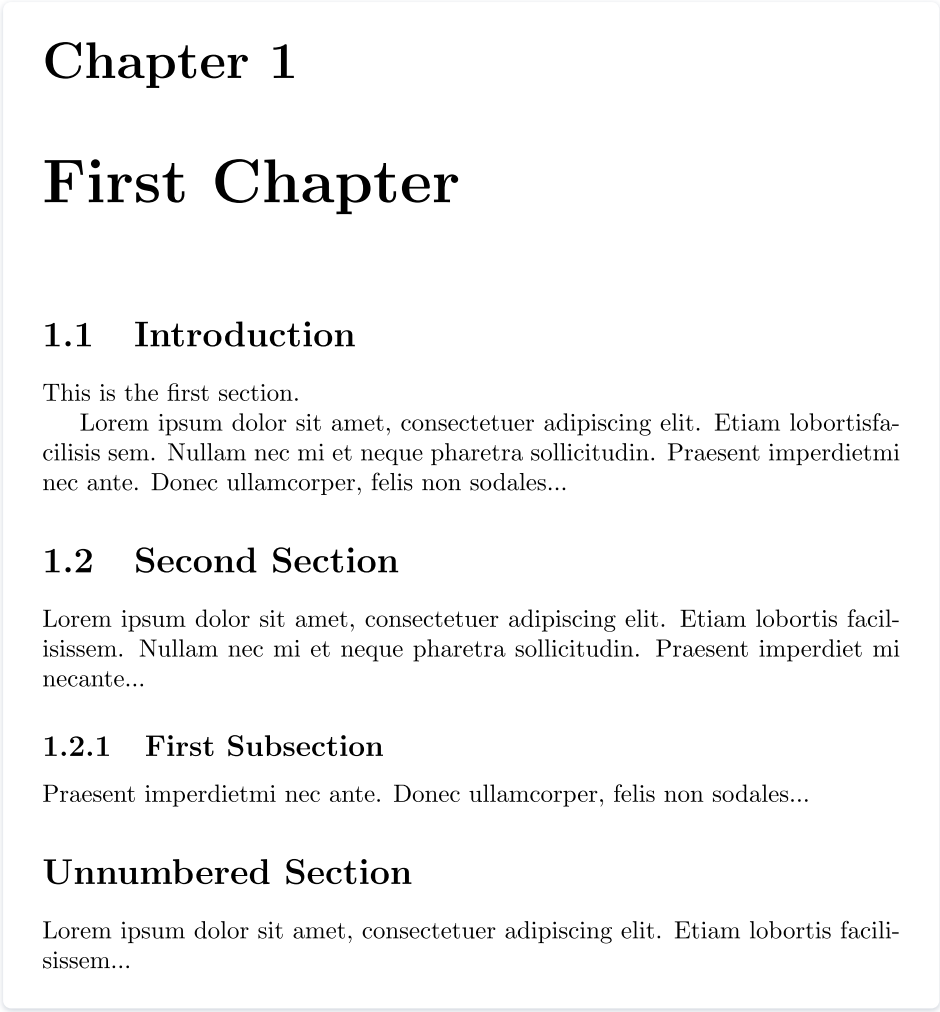
\section*{Unnumbered Section}
Lorem ipsum dolor sit amet, consectetuer
adipiscing elit.
Etiam lobortis facilisissem...
\end{document}
```

```
\文档类{书}\开始{文档}\章{第一章}\节{导言}这是第一节。
Lorem ipsum dolor sit amet，consecetuer
adipiscing elit。Etiam loortisisis sem. Nullam
nec mi et neque pharetra sollicitudin.Praesent
perdietmi nec ante.Donec ullamcorper，felis non
sodales…\节{第二节}Lorem ipsum dolor sit amet，
consecuer adipiscing elit.Etiam loortis
facilisim.Nullam nec mi et neque pharetra
sollicitudin.Praesent无理我无理…\小节{第一小
节}Praesent perdietmi nec ante.Donec
ullamcorper，felis non sodales…\节*{未编号部
分}Lorem ipsum dolor sit amet，consecetuer
adipiscing elit.Etiam loortis facili
```

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：



Chapter 1

First Chapter

1.1 Introduction

This is the first section.

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortisfacilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdietmi nec ante. Donec ullamcorper, felis non sodales...

1.2 Second Section

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisissem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi necante...

1.2.1 First Subsection

Praesent imperdietmi nec ante. Donec ullamcorper, felis non sodales...

Unnumbered Section

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisissem...

The names of sectioning commands are mostly self-explanatory; for example, `\chapter{First Chapter}` creates a new chapter titled `First Chapter`, `\section{Introduction}` produces a section titled `Introduction`, and so forth. Sections can be further divided into `\subsection{...}` and even `\subsubsection{...}`. The numbering of sections, subsections etc. is automatic but can be disabled by using the so-called *starred version* of the appropriate command which has an asterisk (*) at the end, such as `\section*{...}` and `\subsection*{...}`.

章节命令的名称大多一目了然；例如，`\chapter{First Chapter}` 会创建一个标题为 `First Chapter` 的新章节，`\section{Introduction}` 会生成一个标题为 `Introduction` 的小节，以此类推。小节还可以进一步划分为 `\subsection{...}` 甚至 `\subsubsection{...}`。章节、小节等的编号是自动生成的，但可以通过使用相应命令的所谓 *带星号版本* 来禁用，这类命令末尾带有星号（*），比如 `\section*{...}` 和 `\subsection*{...}`。

Collectively, LaTeX document classes provide the following sectioning commands, with specific classes each supporting a

relevant subset:

总体而言，LaTeX 文档类提供了以下分节命令，各类具体文档类分别支持其中的相关子集：

- `\part{part}`
- `\chapter{chapter}`
- `\section{section}`
- `\subsection{subsection}`
- `\subsubsection{subsubsection}`
- `\paragraph{paragraph}`
- `\subparagraph{subparagraph}`

In particular, the `\part` and `\chapter` commands are only available in the `report` and `book` document classes.

特别是，`\part` 和 `\chapter` 命令仅在 `report` 和 `book` 文档类中可用。

Visit the Overleaf article article about sections and chapters for further information about document-structure commands.

请访问 Overleaf 上的关于章节的文章，以获取有关文档结构命令的更多信息。

Creating tables 创建表格

Overleaf provides three options to create tables:

Overleaf 提供了三种创建表格的方法：

1. Using the **Insert Table** button in the **Visual Editor** (or **Code Editor**) toolbar.

   在可视化编辑器（或代码编辑器）的工具栏中使用插入表格按钮。

2. Copying and pasting a table from another document while using **Visual Editor**.

   在可视化编辑器中复制并粘贴表格来自其他文档。

3. Writing the LaTeX code for the table in Code Editor.

   在代码编辑器中编写表格的LaTeX代码。

If you're new to LaTeX, using the toolbar in **Visual Editor** (option 1) is a great way to get started—you can switch between **Visual Editor** and **Code Editor** to see the code

behind the table.

如果你是 LaTeX 新手，使用可视化编辑器中的工具栏（方法1）是入门的绝佳方式——你可以在可视化编辑器和代码编辑器之间切换，查看表格背后的代码。

Here, we focus on option 3—the most flexible method for creating tables—and provide examples that demonstrate how to create tables in LaTeX, including the addition of lines (rules) and captions. As you gain experience, take a look at our detailed guidance on how to create tables using LaTeX.

在这里，我们重点介绍选项3——这是创建表格最灵活的方法，并提供示例来演示如何在LaTeX中创建表格，包括线条（规则）和标题的添加。随着你积累经验，可以查看我们关于如何使用LaTeX创建表格的详细指南。

Creating a basic table in LaTeX
在 LaTeX 中创建基础表格

We start with an example showing how to typeset a basic table:

我们从一个示例开始，展示如何排版一个基础表格：

```
\begin{center}
\begin{tabular}{c c c}
 cell1 & cell2 & cell3 \\
 cell4 & cell5 & cell6 \\
 cell7 & cell8 & cell9
\end{tabular}
\end{center}
```

```
\begin{center}\begin{tabular}{c c c}单元格1 & 单元格2 & 单元格3 \\ 单元格4 & 单元格5 & 单元格6 \\ 单元格7 & 单元格8 & 单元格9
\end{tabular}\end{center}
```

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：

cell1	cell2	cell3
cell4	cell5	cell6
cell7	cell8	cell9

The `tabular` environment is the default LaTeX method to create tables. You must specify a parameter to this environment, in this case `{c c c}` which advises LaTeX that there will be three columns and the text inside each one must be centred. You can also use `r` to right-align the text and `l` to left-align it. The alignment symbol `&` is used to demarcate individual table cells within a table row. To end a table row use the *new line* command `\\`. Our table is contained within a `center` environment to make it centred within the text width of the page.

`tabular`环境是创建表格的默认 LaTeX 方法。你必须为该环境指定一个参数，本例中为 `{c c c}`，它会告知 LaTeX 表格有三列，且每列内的文本需居中显示。你也可以使用 `r` 实现文本右对齐、`l` 实现文本左对齐。对齐符号 `&` 用于分隔表格行内的各个单元格。换行命令 `\\` 用于结束表格行。我们的表格被置于 `center` 环境中，以使其在页面的文本宽度内居中显示。

Adding borders 添加边框

The `tabular` environment supports horizontal and vertical lines (rules) as part of the table:

`tabular`环境支持将水平和垂直线（规则）作为表格的一部分：

- to add horizontal rules, above and below rows, use the `\hline` command

  要在表格行的上方和下方添加水平线，请使用 `\hline` 命令

- to add vertical rules, between columns, use the vertical line parameter `|`

  要在列之间添加竖线，请使用竖线参数`|`

In this example the argument is `{|c|c|c|}` which declares three (centred) columns each separated by a vertical line (rule); in addition, we use `\hline` to place a horizontal rule above the first row and below the final row:

在这个示例中，参数为 `{|c|c|c|}`，它定义了三列（居中），每列之间用竖线（分隔线）隔开；此外，我们使用 `\hline` 在第一行上

方和最后一行下方添加水平分隔线：

```
\begin{center}
\begin{tabular}{|c|c|c|}
 \hline
 cell1 & cell2 & cell3 \\
 cell4 & cell5 & cell6 \\
 cell7 & cell8 & cell9 \\
 \hline
\end{tabular}
\end{center}
```

```
\begin{center}\begin{tabular}{|c|c|c|} \hline单元格1 & 单元格2 & 单元格3 \\ 单元格4 & 单元格5 & 单元格6 \\ 单元格7 & 单元格8 & 单元格9 \\ \hline\end{tabular}\end{center}
```

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：

cell1	cell2	cell3
cell4	cell5	cell6
cell7	cell8	cell9

Here is a further example: 再举一个例子：

```
\begin{center}
 \begin{tabular}{||c c c c||}
 \hline
 Col1 & Col2 & Col2 & Col3 \\ [0.5ex]
 \hline\hline
 1 & 6 & 87837 & 787 \\
 \hline
 2 & 7 & 78 & 5415 \\
 \hline
 3 & 545 & 778 & 7507 \\
 \hline
 4 & 545 & 18744 & 7560 \\
 \hline
 5 & 88 & 788 & 6344 \\ [1ex]
 \hline
```

```
\end{tabular}
\end{center}
```

```
\begin{center}\begin{tabular}{||c c c c||}
\hline列1 & 列2 & 列2 & 列3 \\ [0.5ex]
\hline\hline1 & 6 & 87837 & 787 \\ \hline2 & 7
& 78 & 5415 \\\hline3 & 545 & 778 & 7507
\\\hline4 & 545 & 18744 & 7560 \\\hline5 & 88 &
788 & 6344 \\ [1ex]
\hline\end{tabular}\end{center}
```

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：

Col1	Col2	Col2	Col3
1	6	87837	787
2	7	78	5415
3	545	778	7507
4	545	18744	7560
5	88	788	6344

Captions, labels and references 标题、标签与引用

You can caption and reference tables in much the same way as images. The only difference is that instead of the **`figure`** environment, you use the **`table`** environment.

你可以像处理图片一样为表格添加标题和引用。唯一的区别是，你需要使用 `table` 环境，而不是 `figure` 环境。

```
Table \ref{table:data} shows how to add a table
caption and reference a table.
\begin{table}[h!]
\centering
\begin{tabular}{||c c c c||}
 \hline
 Col1 & Col2 & Col2 & Col3 \\ [0.5ex]
 \hline\hline
 1 & 6 & 87837 & 787 \\
 2 & 7 & 78 & 5415 \\
 3 & 545 & 778 & 7507 \\
```

```
 4 & 545 & 18744 & 7560 \\
 5 & 88 & 788 & 6344 \\ [1ex]
 \hline
\end{tabular}
\caption{Table to test captions and labels.}
\label{table:data}
\end{table}
```

表\ref{table:data}展示了如何添加表格标题以及引用表格。\begin{table}[h!]\centering\begin{tabular}{||c c c c||} \hline列1 & 列2 & 列2 & 列3 \\ [0.5ex] \hline\hline1 & 6 & 87837 & 787 \\ 2 & 7 & 78 & 5415 \\3 & 545 & 778 & 7507 \\4 & 545 & 18744 & 7560 \\5 & 88 & 788 & 6344 \\ [1ex] \hline\end{tabular}\caption{测试标题和标签的表格。}\label{table:data}\end{table}

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：

Table 1 shows how to add a table caption and reference a table.

Col1	Col2	Col2	Col3
1	6	87837	787
2	7	78	5415
3	545	778	7507
4	545	18744	7560
5	88	788	6344

Table 1: Table to test captions and labels.

Adding a Table of Contents 添加目录

Creating a table of contents is straightforward because the command `\tableofcontents` does almost all the work for you:

创建目录非常简单，因为命令 `\tableofcontents` 几乎为你完成了所有工作：

```
\documentclass{article}
\title{Sections and Chapters}
\author{Gubert Farnsworth}
\date{August 2022}
```

```
\begin{document}

\maketitle

\tableofcontents

\section{Introduction}

This is the first section.

Lorem ipsum dolor sit amet, consectetuer adipiscing
elit. Etiam lobortisfacilisis sem. Nullam nec mi et
neque pharetra sollicitudin. Praesent imperdietmi nec ante.
Donec ullamcorper, felis non sodales...

\section*{Unnumbered Section}
\addcontentsline{toc}{section}{Unnumbered Section}

Lorem ipsum dolor sit amet, consectetuer adipiscing elit.
Etiam lobortis facilisissem. Nullam nec mi et neque pharetra
sollicitudin. Praesent imperdiet mi necante...

\section{Second Section}

Lorem ipsum dolor sit amet, consectetuer adipiscing elit.
Etiam lobortis facilisissem. Nullam nec mi et neque pharetra
sollicitudin. Praesent imperdiet mi necante...
\end{document}
```

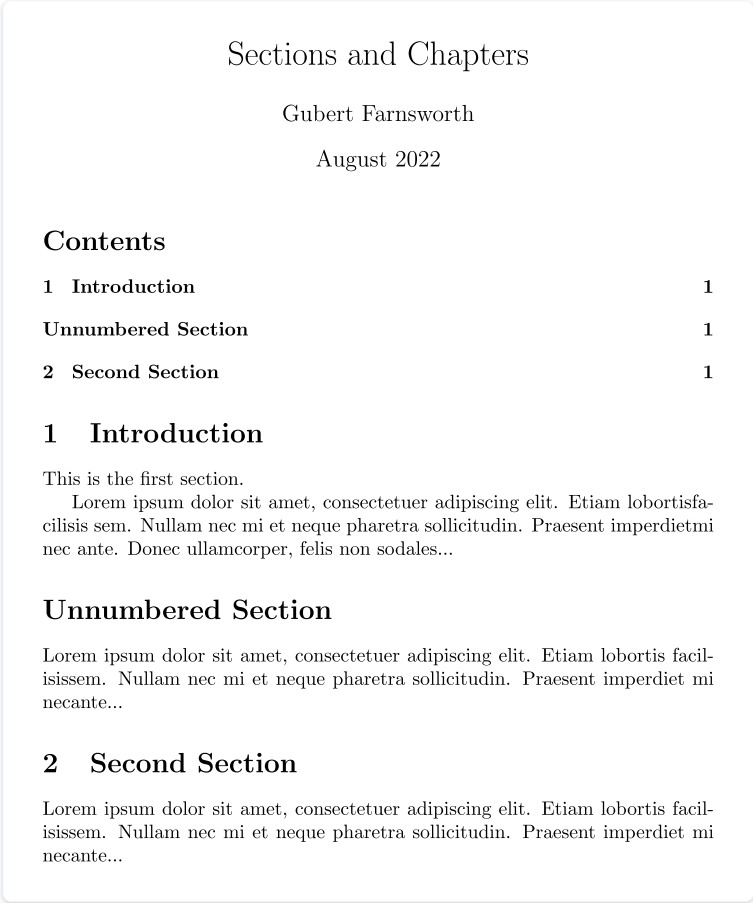
```
\docentclass{文章}\标题{章节和章节}\作者{古伯特·法恩斯沃斯}\日期{2022年8月}\开始{文档}\maketitle\tableofcontent\部分{简介}这是第一部分。Lorem ipsum dolor sit amet，consecetuer adipiscing elit。Etiam loortisisis sem. Nullam nec mi et neque pharetra sollicitudin.Praesent impdietmi nec ante.Donulec lamcorper，felis non
```

```
sodales…\部分*{未编号部分}\addcontentsline{toc}{部分}{未编号部分}Lorem ipsum dolor sit amet, consecuer adipiscing elit.Etiam loortis facilisim.Nullam mi et neque pharetra sollicitudin.Praesent permi ecante…\部分{第二部分}Lorem ipsum dolor sit amet, consecetuer adipet
```

Open this example in Overleaf. 在 Overleaf 中打开此示例。

This example produces the following output:

该示例生成以下输出：



Sections and Chapters

Gubert Farnsworth

August 2022

Contents

1 Introduction 1

Unnumbered Section 1

2 Second Section 1

1 Introduction

This is the first section.

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortisfacilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdietmi nec ante. Donec ullamcorper, felis non sodales...

Unnumbered Section

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisissem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi necante...

2 Second Section

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisissem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi necante...

Sections, subsections and chapters are automatically included in the table of contents. To manually add entries, such as an unnumbered section, use the command `\addcontentsline` as shown in the example.

章节、小节和子章节会自动收录到目录中。若要手动添加条目（如未编号的小节），请使用命令 `\addcontentsline`，示例如下。

Downloading your finished document
下载完成的文档

The following brief video clip shows how to download your project's source code or the typeset PDF file:

下面这段简短的视频片段展示了如何下载项目的源代码或排版后的 PDF 文件：



More information can be found in the Overleaf help article Exporting your work from Overleaf.

更多信息可查看 Overleaf 帮助文章 从 Overleaf 导出你的作品。

Finding and using LaTeX packages
查找并使用 LaTeX 宏包

LaTeX not only delivers significant typesetting capabilities but also provides a *framework for extensibility* through the use of add-on *packages*. Rather than attempting to provide commands and features that "try to do everything", LaTeX is designed to be *extensible*, allowing users to load external bodies of code (packages) that provide more specialist typesetting capabilities or extend LaTeX's built-in features—such as typesetting tables. As observed in the section Adding images, the `graphicx` package extends LaTeX by providing commands to import graphics files and was loaded (in the preamble) by writing

LaTeX 不仅具备强大的排版功能，还通过附加*宏包*提供了*可扩展框架*。LaTeX 并非试图打造"无所不能"的命令与功能，而是被设计为*可扩展*的系统，允许用户加载外部代码体（宏包），以实现更专业的排版能力或扩展 LaTeX 内置功能——比如表格排版。正如添加图片一节所述，`graphicx`宏包通过提供导入图形文件的命令扩展了 LaTeX 功能，用户需在导言区通过以下方式加载该宏包：

```
\usepackage{graphicx}
```

Loading packages 加载宏包

As noted above, packages are loaded in the document preamble via the `\usepackage` command but because (many) LaTeX packages provide a set of *options*, which can be used to configure their behaviour, the `\usepackage` command often looks like this:

如上所述，宏包是通过`\usepackage`命令在文档导言区加载的，但由于（许多）LaTeX 宏包提供了一组*选项*，可用于配置其行为，因此`\usepackage`命令通常如下所示：

```
\usepackage[options]{somepackage}
\usepackage[选项]{某个宏包}
```

The square brackets "`[...]`" inform LaTeX which set of options should be applied when it loads `somepackage`. Within the set of options requested by the user, individual options, or settings, are typically separated by a comma; for example, the `geometry` package provides many options to configure page layout in LaTeX, so a typical use of `geometry` might look like this:

方括号"`[...]`"会告知 LaTeX 在加载 `somepackage` 时应应用哪一组选项。在用户请求的选项集合中，单个选项或设置通常用逗号分隔；例如，`geometry` 宏包提供了众多配置 LaTeX 页面布局的选项，因此 `geometry` 宏包的典型用法如下所示：

```
\usepackage[total={6.5in,8.75in},
top=1.2in, left=0.9in, includefoot]{geometry}
\usepackage[总宽度=6.5英寸,总高度=8.75英寸,顶部边距
=1.2英寸,左侧边距=0.9英寸,包含页脚]{geometry}
```

The `geometry` package is one example of a package written and contributed by members of the global LaTeX community and made available, for free, to anyone who wants to use it.

`geometry`宏包是由全球 LaTeX 社区成员编写并贡献的宏包之一，任何有使用需求的人都可以免费获取。

If a LaTeX package does not provide any options, or the user wants to use the default values of a package's options, it would be loaded like this:

如果某个 LaTeX 宏包不提供任何选项，或者用户希望使用宏包选项的默认值，就可以按以下方式加载它：

```
\usepackage{somepackage}
```

When you write `\usepackage[...]{somepackage}` LaTeX looks for a corresponding file called *somepackage*`.sty`, which it needs to load and process—to make the package commands available and execute any other code provided by that package. If LaTeX cannot find *somepackage*`.sty` it will terminate with an error, as demonstrated in the following Overleaf example:

当你编写 `\usepackage[...]{somepackage}` 时，LaTeX 会查找一个名为 `somepackage.sty` 的对应文件，它需要加载并处理该文件——以使宏包命令可用，并执行该宏包提供的其他代码。如果 LaTeX 找不到 `somepackage.sty`，它会终止并报错，如下方的 Overleaf 示例所示：

```
\documentclass[12pt, letterpaper]{article}
\usepackage{somepackage}% a NON-EXISTENT
package
\begin{document}
This will fail!
\end{document}
```

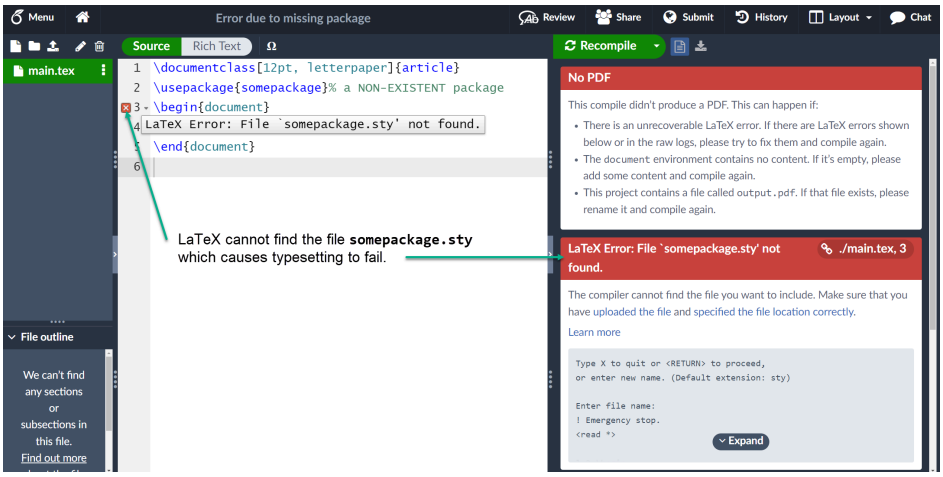
```
\documentclass[12pt, letterpaper]
{article}\usepackage{somepackage}% 一个不存在的宏包\begin{document}这会编译失败！ \end{document}
```

Open this ***error-generating example*** on Overleaf

在 Overleaf 上打开这个 ***生成错误的示例***

This example produces the following output:

该示例生成以下输出：



Finding information about packages: CTAN

查找有关宏包的信息：CTAN

Packages are distributed through the Comprehensive TeX Archive Network, usually referred to as CTAN, which, at the time of writing, hosts 6287 packages from 2881 contributors. CTAN describes itself as

软件包通过TeX 综合档案网络分发，通常被称为 CTAN，在撰写本文时，该网络托管着来自 2881 位贡献者的 6287 个软件包。CTAN 将自己描述为

> ... a set of Internet sites around the world that offer TEX-related material for download.
>
> ……是全球各地提供与TeX相关的可下载资料的一系列网站。

You can browse CTAN to look for useful packages; for example:

你可以浏览 CTAN 寻找有用的包，例如：

- by topic 按主题分类
- alphabetically (useful if you know the package name)

  按字母顺序（如果你知道宏包名称，此方式很实用）

You can also use the search facility (at the top of the page).

你也可以使用搜索工具（在页面顶部）。

Packages available on Overleaf: Introducing TeX Live

Overleaf 上可用的宏包：TeX Live 简介

Once per year a (large) *subset* of packages hosted on CTAN, plus LaTeX-related fonts and other software, is collated and

distributed as a system called TeX Live, which can be used to install your own (local) LaTeX setup. In fact, Overleaf's servers also use TeX Live and are updated when a new version of TeX Live is released. Overleaf's TeX Live updates are not immediate but take place a few months post-release, giving us time to perform compatibility tests of the new TeX Live version with the thousands of templates contained in our gallery. For example, here is our TeX Live 2025 upgrade announcement.

CTAN 上托管的（大型）*子集*宏包、与 LaTeX 相关的字体以及其他软件，每年会整理一次，并以名为 TeX Live 的系统形式发布，你可通过该系统安装本地的 LaTeX 环境。事实上，Overleaf 的服务器也使用 TeX Live，且每当 TeX Live 发布新版本时，服务器都会随之更新。Overleaf 对 TeX Live 的更新并非即时完成，而是会在新版本发布数月后进行，以便我们有时间对新版本 TeX Live 与图库中数以千计的模板进行兼容性测试。例如，你可以查看我们的 TeX Live 2025 升级公告。

Although TeX Live contains a (large) *subset* of CTAN packages it is possible to find an interesting package, such as `igo` for typesetting Go diagrams, which is hosted on CTAN but not included in (distributed by) TeX Live and thus unavailable on Overleaf. Some packages hosted on CTAN are not part of TeX Live due to a variety of reasons: perhaps a package is obsolete, has licensing problems, is extremely new (recently uploaded) or has platform dependencies, such as working on Windows but not Linux.

尽管TeX Live包含了CTAN宏包的一个（庞大的）*子集*，但仍有可能找到一些有趣的宏包，比如用于排版围棋棋谱的`igo`宏包，这类宏包托管在CTAN上，却未被TeX Live收录（由其分发），因此在Overleaf上无法使用。部分托管在CTAN上的宏包未纳入TeX Live的原因多种多样：可能是宏包已过时、存在授权问题、刚发布（最近上传），或是存在平台依赖问题，例如仅支持Windows系统，不支持Linux系统。

New packages, and updates to existing ones, are uploaded to CTAN all year round but updates to TeX Live are distributed annually; consequently, packages contained in the current version of TeX Live will not be as up-to-date as those hosted on CTAN. Because Overleaf's servers use TeX Live it is possible that packages installed on our servers—i.e., ones available to our users—might not be the very latest versions available on

CTAN but, generally, this is unlikely to be problematic.

新的宏包以及现有宏包的更新会全年上传至CTAN，但TeX Live的更新则按年度发布；因此，TeX Live当前版本中包含的宏包不会像CTAN上托管的宏包那样保持最新状态。由于Overleaf的服务器使用TeX Live，我们服务器上安装的宏包——即用户可使用的宏包——可能并非CTAN上的最新版本，但通常情况下这不会造成问题。

Documentation Home 文档首页

Overleaf Learning Center Overleaf 学习中心

Overleaf guides Overleaf 指南

Overleaf user documentation Overleaf 用户文档

Overleaf group administration Overleaf 团队管理

Overleaf Commons administration Overleaf 共享库管理

Overleaf on-premises Overleaf 本地部署版

LaTeX basics LaTeX 基础

Creating your first LaTeX document 创建你的第一个 LaTeX 文档

Paragraphs and new lines 段落与换行

Bold, italics and underlining 粗体、斜体和下划线

Lists 列表

Errors 错误

Mathematics 数学

Mathematical expressions 数学表达式

Subscripts and superscripts 下标和上标

Brackets and Parentheses 方括号与圆括号

Matrices 矩阵

Fractions and Binomials 分式与二项式

Aligning equations 方程对齐

Operators 运算符

Spacing in math mode 数学模式中的间距

Integrals, sums and limits 积分、求和与极限

Display style in math mode 数学模式中的显示样式

List of Greek letters and math symbols

希腊字母与数学符号列表

Mathematical fonts 数学字体

Using the Symbol Palette in Overleaf

在 Overleaf 中使用符号面板

Figures and tables 数字和表格

Tables 桌子

Positioning Images and Tables 图片与表格的排版

Lists of Tables and Figures 表格与图表目录

Drawing Diagrams Directly in LaTeX 直接在 LaTeX 中绘制图表

TikZ package TikZ 宏包

Document structure 文档结构

Sections and chapters 节与章

Table of contents 目录

Cross referencing sections, equations and floats

交叉引用章节、公式和浮动体

Indices 索引

Glossaries 术语表

Nomenclatures 命名法

Management in a large project 大型项目中的管理

Multi-file LaTeX projects 多文件LaTeX项目

Hyperlinks 超链接

Formatting 格式设置

Lengths in LaTeX LaTeX 中的长度

Headers and footers 页眉和页脚

Page numbering 页码

Paragraph formatting 段落格式

Line breaks and blank spaces 换行符和空格

Text alignment 文本对齐

Page size and margins 页面大小和边距

Single sided and double sided documents

单面和双面文件

Multiple columns 多列

Counters 计数器

Code listing 代码列表

Code Highlighting with minted 使用 minted 进行代码高亮

Using colours in LaTeX 在 LaTeX 中使用颜色

Footnotes 脚注

Margin notes 边注

References and citations 参考文献与引用

Bibliography management with bibtex 使用bibtex管理参考文献

Bibliography management with natbib 使用 natbib 管理参考文献

Bibliography management with biblatex

使用biblatex管理参考文献

Bibtex bibliography styles Bibtex 参考文献样式

Natbib bibliography styles Natbib 参考文献样式

Natbib citation styles Natbib 引用样式

Biblatex bibliography styles Biblatex 参考文献格式

Biblatex citation styles Biblatex 引用样式

Fonts 字体

Font sizes, families, and styles 字体大小、字族和字体样式

Font typefaces 字体样式

Supporting modern fonts with XeLaTeX

使用 XeLaTeX 支持现代字体

Presentations 演示文稿

Beamer 贝美

Powerdot

Posters 海报

Commands 命令

Commands 命令

Environments 环境

Languages 语言

Multilingual typesetting on Overleaf using polyglossia and fontspec

在 Overleaf 上使用 polyglossia 和 fontspec 进行多语言排版

Multilingual typesetting on Overleaf using babel and fontspec

在 Overleaf 上使用 babel 和 fontspec 进行多语言排版

International language support 国际语言支持

Quotations and quotation marks 引号和引号标记

Arabic 阿拉伯语

Chinese 中文

French 法语

German 德语

Greek 希腊语

Italian 意大利语

Japanese 日语

Korean 韩语

Portuguese 葡萄牙语

Russian 俄语

Spanish 西班牙语

Field specific 特定领域

Theorems and proofs 定理与证明

Chemistry formulae 化学公式

Feynman diagrams 费恩曼图

Molecular orbital diagrams 分子轨道图

Chess notation 国际象棋记谱法

Knitting patterns 编织图案

CircuiTikz package CircuiTikz 宏包

Pgfplots package Pgfplots 绘图包

Typesetting exams in LaTeX 用LaTeX排版考试试卷

Attribute Value Matrices 属性值矩阵

Class files 类文件

Understanding packages and class files

了解宏包和类文件

List of packages and class files 宏包与类文件列表

Writing your own package 编写你自己的宏包

Writing your own class 编写你自己的文档类

Advanced TeX/LaTeX 高级 TeX/LaTeX

In-depth technical articles on TeX/LaTeX

关于TeX/LaTeX的深度技术文章